Title of the Project	**STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students**
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Rationale	Physical inactivity has been identified as one of the leading risk factors for noncommunicable diseases worldwide. According to the World Health Organization (WHO, 2023), insufficient physical activity increases the risk of cardiovascular diseases, diabetes, obesity, and mental health disorders. In the Philippines, the Department of Health (DOH) adopted the Philippine Physical Activity Guidelines (PPAG), which recommend that adults engage in at least 150–300 minutes of moderate-intensity physical activity per week, while children and adolescents should accumulate at least 60 minutes of physical activity daily (Department of Health [DOH], 2012). Research translating these time-based recommendations into step equivalents suggests that approximately 7,000–8,000 steps per day are associated with significant health benefits and reduced mortality risk among adults (Paluch et al., 2021), while 8,000–10,000 steps per day are commonly aligned with meeting moderate-to-vigorous activity thresholds (Tudor-Locke et al., 2020). In the Philippine context, physical inactivity among youth remains a significant concern. The 2022 Philippine Report Card on Physical Activity for Children and Adolescents reported that a large proportion of Filipino children and adolescents do not meet recommended physical activity levels, resulting in low grades for overall physical activity indicators (Aubert et al., 2022). These findings suggest that sedentary behavior is prevalent at early ages and may continue into tertiary education. Among university students, academic workload, prolonged screen time, and technology-dependent lifestyles contribute to reduced physical activity levels. Research indicates that physical activity is positively associated with improved well-being and emotional engagement among Filipino tertiary students (Liang et al., 2023). This suggests that insufficient physical movement may negatively affect both physical and mental health outcomes within the student population. Student wellness and mental health promotion are also recognized priorities in Philippine higher education institutions. The Commission on Higher Education (CHED, 2018) issued policies and guidelines on mental health in higher education institutions to strengthen student support systems and promote preventive wellness initiatives. Encouraging regular physical activity aligns with these national efforts to improve student well-being. Despite the availability of numerous fitness and step-tracking applications, research suggests that many applications fail to sustain long-term engagement because they rely primarily on numerical step feedback without incorporating motivational and interactive features (Yang et al., 2021). Gamification has been shown to significantly improve

	motivation and participation in physical activity programs by integrating rewards, goal-setting, and challenges (Mazeas et al., 2022). Additionally, augmented reality (AR) technologies have demonstrated potential in enhancing user engagement and enjoyment in movement-based activities among young adults (Lu et al., 2025). However, there remains a lack of locally contextualized mobile applications in the Philippines that integrate augmented reality, gamification, and structured health-aligned step systems within a university campus setting. Most existing applications are designed for general populations and do not tailor engagement strategies to campus environments or align goals with recommended physical activity classifications. This gap highlights the need for a student-centered, campus-based mobile application that combines step tracking, gamification, and augmented reality to promote sustained physical activity among university students. Therefore, this study proposes the development of STEP-UP: An Augmented Reality Mobile Application for Promoting Physical Activity Among University Students, which aims to provide a context-specific and engaging technological solution that encourages daily movement and supports healthier lifestyle behaviors.
Objectives	**General Objective** To develop and evaluate an augmented reality mobile application (STEP-UP) designed to increase step count and improve physical activity behavior among university students. • To develop a mobile application that integrates automatic step tracking, augmented reality walking mode, gamification features, and health guidance tools. • To determine the baseline average daily step count of selected university students prior to using the STEP-UP application. • To compare the baseline and post-intervention average daily step count to determine whether there is a significant increase after using the STEP-UP application. • To assess changes in physical activity behavior and motivation after the intervention period. • To evaluate the quality, user engagement, and motivational effectiveness of the STEP-UP application using ISO/IEC 25010 software quality standards and survey instruments adapted from established gamification and behavioral frameworks.

Review of Related Literatures

Physical Inactivity and Sedentary Behavior Among University Students

Physical inactivity and sedentary behavior are common among university students and are considered modifiable risk factors for an unhealthy lifestyle. Edelmann et al. (2022) highlighted that university students can show unhealthy patterns of physical activity and sedentary behavior, supporting the need for interventions that target student lifestyle behaviors. Additionally, interventions aimed at improving physical activity in university students have been found to increase total physical activity and step counts, indicating that behavior-focused programs can make measurable improvements in student activity levels (Yuan et al., 2024).

Physical Inactivity and Sedentary Behavior in the Philippines

Physical inactivity is recognized as one of the leading risk factors for noncommunicable diseases worldwide. According to the World Health Organization (WHO, 2023), insufficient physical activity increases the risk of cardiovascular diseases, diabetes, obesity, and poor mental health outcomes. The WHO recommends that adults engage in at least 150 minutes of moderate-intensity physical activity per week, while adolescents should accumulate at least 60 minutes per day.

In the Philippine context, physical inactivity among children and adolescents remains a major concern. The 2022 Philippine Report Card on Physical Activity for Children and Adolescents reported that a large proportion of Filipino youth do not meet recommended physical activity levels, resulting in low grades for overall physical activity indicators (Aubert et al., 2022). The report highlights widespread sedentary behavior and limited participation in organized physical activities.

Physical Activity and Mental Health Among Students

Physical activity is strongly associated with improved mental health outcomes. Regular physical activity has been linked to reduced symptoms of anxiety and depression, as well as improved psychological well-being (World Health Organization [WHO], 2023).
In higher education settings, student wellness has become a growing concern. The Commission on Higher Education (CHED) emphasizes the importance of student affairs and wellness services in promoting holistic student development. Encouraging physical activity within campus environments aligns with national policies that aim to improve student well-being.

Furthermore, research among Filipino tertiary students indicates that positive health behaviors, including regular physical activity, significantly predict improved well-being and emotional engagement (Frontiers in Psychology, 2025). These findings reinforce the importance of integrating physical activity promotion into university-based interventions.

Limitations of Traditional Fitness Apps and the Need for Engagement

While fitness and step-tracking apps are widely available, sustained use is strongly affected by user motivation and engagement. Yang et al. (2021) found that intentions to use fitness apps and intentions to be physically active are influenced by determinants such as usefulness and features related to education, motivation, and gamification implying that apps must be designed to support goal achievement and habit formation to remain effective. These findings support the need for a more engaging approach (beyond basic step counting) for university students consistent with STEP-UP's AR + gamification focus.

Gamification for Promoting Physical Activity

Gamification is widely recognized as an effective strategy to encourage physical activity by increasing motivation, enjoyment, and adherence. A systematic review and meta-analysis by Mazeas et al. (2022) concluded that gamified interventions are promising for promoting physical activity and can remain effective beyond the initial novelty effect. In addition, research examining health and well-being apps suggests that behavior change features (e.g., feedback, goal-setting, and rewards) are important components that can improve the potential of apps to support healthier behaviors (McKay et al., 2019). These findings directly support STEP-UP's features such as step-to-point conversion, missions, goals, and avatar rewards.

Augmented Reality and Exergaming to Increase Activity and Enjoyment

Augmented reality (AR) and exergaming have gained attention for making movement-based activities more immersive and enjoyable. Philippe et al. (2024) examined an AR exergame in young adults and discussed AR exergaming as a way to support aerobic training, reinforcing the potential of AR-based activity systems for populations similar in age to university students. Moreover, active augmented reality (AAR) games have been studied for their potential to increase physical activity in safe and structured ways (Lu et al., 2025). These studies support STEP-UP's AR walking view (camera-based overlay) as a mechanism to increase engagement during real-world walking.

Location-Based AR Games as Evidence That AR Can Promote Walking

Location-based AR games provide real-world evidence that game design can increase walking behavior by linking progress to movement. A systematic literature review reported that Pokémon GO showed positive immediate effects on physical health such as increased steps and distance moved, though benefits may depend on continued play (Wang & Wang, 2021). Related work also reviewed Pokémon GO's influence on physical activity and psychosocial outcomes, indicating that AR-driven gameplay can motivate movement through interactive goals and exploration (Liang et al., 2023). This supports STEP-UP's concept of using game mechanics and interactive experience to encourage walking among students.

Behavior Change Techniques and Persuasive Features in Health Apps

Behavior change techniques (BCTs) are commonly used to improve the effectiveness of health applications. Milne-Ives et al. (2020) systematically reviewed mobile apps for health behavior change and examined the inclusion of BCTs—supporting the importance of features like goal setting, feedback, reminders, and progress tracking in behavior-focused apps. Similarly, Edwards et al. (2016) systematically reviewed health apps with gaming elements and analyzed embedded BCTs, reinforcing that gamified app features can be designed to actively support behavior change. This aligns with STEP-UP's health guidance features (posture tutorial, warm-up/cooldown guide, hydration reminders, and health tips) as supportive tools that complement gamified walking.

Step Tracking Accuracy and Data Integrity (Relevant to Anti-Cheat / Vehicle Detection)

Accurate step measurement is essential in step-based applications because points, rewards, and progress rely on reliable tracking. Yao et al. (2022) evaluated step

	accuracy and reliability of smartphone-based step tracking compared with an accelerometer in free-living conditions, highlighting that step counting accuracy and reliability are important considerations when using smartphone sensors for activity monitoring. These findings support the inclusion of STEP-UP's anti-cheat feature (vehicle detection) to reduce inaccurate step counting and help maintain fairness and reliability in the gamified reward system.
Methodology	**Research Design** This study will employ a design and developmental research approach combined with a quasi-experimental one-group pretest–posttest design. The developmental component focuses on the planning, design, development, and implementation of the STEP-UP mobile application, integrating step tracking, augmented reality, gamification mechanics, and health guidance features. The quasi-experimental component will assess the effectiveness of the developed application in increasing average daily step count and improving physical activity behavior among students. A one-group pretest–posttest design will be used to compare participants' baseline step counts prior to application use with post-intervention step counts after using the STEP-UP application. . **System Development Life Cycle (SDLC)** This study will use the **Agile Software Development Life Cycle (SDLC)** as the development model for the STEP-UP mobile application. Agile SDLC is an iterative and incremental approach that will allow the system to be developed in stages, with each stage including planning, design, development, testing, and evaluation. This approach will allow continuous improvement of the system based on testing results and feedback. The Agile SDLC will consist of the following phases: **Planning Phase** During this phase, the researchers will identify the system requirements, objectives, and core features of the application. This will include defining system functionalities such as step tracking, augmented reality visualization, gamification, and data synchronization. **Design Phase** In this phase, the system architecture, user interface, and application structure will be designed. The layout of menus, augmented reality interface, and activity tracking displays will be planned to ensure usability and efficiency. **Development Phase** During the development phase, the mobile application will be developed using Unity as the primary development engine, C# as the programming language, and Visual Studio Code as the integrated development environment. Core system features such as step tracking, augmented reality visualization, gamification, and coded speed threshold detection will be implemented. Firebase will be integrated to support cloud data storage and synchronization. **Testing Phase** The system will undergo testing to ensure proper functionality, reliability, and accuracy.

Each feature, including step tracking, augmented reality functionality, speed threshold detection, and data synchronization, will be tested to identify and correct system errors.

Evaluation Phase

After system development and testing, the application will be evaluated by selected respondents and IT experts using ISO/IEC 25010 software quality standards. Feedback will be collected and analyzed to assess the system's usability, functionality, and effectiveness.

Development Tools and Technologies

The system will be developed using the following tools and technologies:

- **Unity Engine** – for mobile application and augmented reality development
- **C# Programming Language** – for system logic and feature implementation
- **Visual Studio Code** – for coding, debugging, and system development
- **Firebase** – for cloud database, data storage, and synchronization
- **Android Mobile Devices** – for system deployment and testing

System Development and Architecture

The proposed system will be developed as a mobile application using Unity as the primary development engine, with C# as the main programming language and Visual Studio Code as the integrated development environment (IDE). Unity will be used to design and implement the application interface, augmented reality functionality, and overall system behavior. C# will be used to develop the application logic, including step tracking, and gamification features.

The application will utilize the smartphone's built-in step counter functionality to monitor user steps in real time. This allows accurate tracking of physical activity without requiring additional hardware devices.

To ensure accurate step tracking, the system will implement a coded speed threshold detection mechanism using C# logic. A predefined speed threshold of 10 kilometers per hour will be set within the system. When the detected speed exceeds this threshold, the step tracking function will automatically pause to prevent inaccurate step counting caused by vehicle movement. Once the detected speed falls below the defined threshold, the system will resume step tracking automatically. This mechanism will help ensure that only legitimate walking activity will be recorded.

The system will also integrate augmented reality functionality using Unity's AR capabilities, allowing virtual elements such as avatars and activity indicators to be displayed through the smartphone camera.

For data storage and synchronization, the system will use Firebase as the cloud database platform. Firebase will allow secure storage of user data, including step records, points, and rewards. It will also support offline and online synchronization, enabling the application to store data locally when internet access is unavailable and automatically synchronize data with the cloud when connectivity is restored.

The system architecture will consist of the following components:

- Unity-based user interface and augmented reality module
- C# application logic for step tracking and gamification
- Local data storage within the mobile device
- Firebase cloud database for data synchronization and backup

Population and Sample Size Determination

The target population of this study consists of students enrolled in the College of Engineering and Computer Studies (COECS) of Pampanga State Agricultural University (PSAU).

The required sample size was determined using statistical power analysis for a paired-samples t-test, as the study aims to detect a significant difference between baseline and post-intervention step counts.

Using:
• Significance level (α) = 0.05
• Statistical power = 0.80
• Assumed moderate effect size (Cohen's d = 0.50), commonly used in behavioral intervention studies

A moderate effect size was assumed due to the behavioral nature of the intervention and the absence of prior local pilot data.

The minimum required sample size for detecting a statistically significant change was estimated at approximately 34 participants.

To account for possible attrition, incomplete step logs, or withdrawal during the intervention period, an additional 15–20% buffer was included. Therefore, the study will target 40 respondents.

This sample size ensures sufficient statistical sensitivity to detect meaningful changes in physical activity levels.

Sampling Technique

This study will utilize purposive sampling in selecting respondents from the College of Engineering and Computer Studies (COECS).
Participants must meet the following criteria:

- Currently enrolled student of COECS
- Own an Android smartphone capable of running the STEP-UP application
- Have a built-in motion sensor or step-tracking capability
- Regularly use smartphones for academic or personal purposes
- Willing to participate in baseline monitoring, intervention usage, and evaluation

Purposive sampling is appropriate because the study requires participants who meet specific technical and usage requirements to ensure accurate system testing and evaluation.

Intervention duration

The study will use a seven (7)-day baseline monitoring period followed by a twenty-one

(21)-day intervention period. The baseline period establishes participants' typical step count prior to exposure to the application. The three-week intervention period allows participants to experience multiple cycles of the application's gameplay mechanics (daily goals and weekly missions) and provides sufficient time to observe changes beyond the initial novelty effect. This duration is appropriate for a pilot, campus-based evaluation of a gamified mobile health intervention while remaining feasible within capstone research constraints.

Data Collection Instruments

Data for this study will be collected using multiple instruments to measure system effectiveness, behavioral change, and software quality.

1. Step Count Monitoring

Baseline step counts will be collected using participants' existing smartphone step-tracking applications (e.g., Google Fit, Samsung Health, or built-in step counters).
During the intervention phase, step counts will be automatically recorded by the STEP-UP application using the smartphone's built-in motion sensors.
Average daily step counts during the baseline and intervention periods will serve as the primary quantitative variable for effectiveness analysis.

2. Physical Activity and Motivation Questionnaire

A structured survey questionnaire will be administered before and after the intervention period to assess changes in physical activity behavior and motivation.

The questionnaire will be adapted from constructs of **Self-Determination Theory (SDT)**, measuring:

- Perceived competence
- Autonomy
- Motivation toward physical activity
- Behavioral intention to remain active

Responses will be measured using a 5-point Likert scale ranging from Strongly Disagree (1) to Strongly Agree (5).

3. Gamification and Gameplay Experience Questionnaire

To evaluate the effectiveness of the application's gameplay mechanics, a survey instrument adapted from established gamification research literature and selected constructs of the Game Experience Questionnaire (GEQ)

The instrument will assess:

- User engagement
- Enjoyment
- Perceived reward effectiveness
- Competitive motivation (leaderboard)
- Immersion in augmented reality mode

Responses will be measured using a 5-point Likert scale.

4. ISO/IEC 25010 Software Quality Evaluation
The application will be evaluated using selected characteristics of the ISO/IEC 25010

software quality model, including:

- Functional suitability
- Usability
- Reliability
- Performance efficiency

This evaluation will be completed by selected respondents after the intervention period.

Application-Based Insight Parameters

In addition to survey responses, system-generated data will be analyzed to gather behavioral insights from the application.
The following parameters will be collected:

- Average daily step count (baseline and intervention)
- Weekly average step count (Week 1, Week 2, Week 3)
- Total accumulated points
- Daily mission completion rate
- Weekly mission completion rate
- Leaderboard participation frequency

These parameters allow objective evaluation of user engagement with gamification mechanics and sustained participation throughout the intervention period.

Data Analysis

Descriptive statistics, including mean, frequency, and percentage, will be used to summarize survey responses and system evaluation results.
To determine whether there is a statistically significant difference between baseline and post-intervention average daily step counts, a **paired-samples t-test** will be conducted.
The level of significance will be set at:
$\alpha = 0.05$
Effect size (Cohen's d) will also be computed to measure the magnitude of change in physical activity levels.
Statistical analysis will be performed using appropriate statistical software. Assumptions of normality will be checked prior to conducting the paired-samples t-test.

Ethical Considerations

This study will comply with institutional research ethics guidelines and the Data Privacy Act of 2012 (RA 10173).
Prior to participation, respondents will be provided with an informed consent form outlining:

- The purpose of the study
- Voluntary participation
- Right to withdraw at any time
- Data confidentiality measures

No sensitive personal information will be collected. User data will be anonymized using participant codes. Step count data and survey responses will be used solely for research purposes.
Cloud synchronization through Firebase will store only necessary activity data and will not include personal identification details.

Expected Output (Proposed Features and Functionality to the Product)	The expected output of this project is a fully functional mobile application named **STEP-UP**, designed to promote physical activity among university students through augmented reality, gamification, and health guidance. The application will include the following features 1. **User Registration and Login System** – The application will allow users to create an account and securely log in to access personalized step tracking, progress, and rewards. User data will be stored using Firebase to ensure data security and continuity. 2. **User-Friendly Main Interface** – The application will have an organized and easy-to-use main menu that allows users to quickly access core features such as Play, Customize, Settings, and Tutorial. The interface will be designed to be simple and intuitive for smooth navigation. 3. **Local Data Storage System** – The system will store user information, step records, earned points, and customization progress locally on the device to ensure accessibility even without an internet connection. 4. **Offline Mode with Online Synchronization** – The application will support offline usage by storing activity data locally. Once internet connectivity is restored, the system will automatically synchronize saved data with the Firebase cloud database. 5. **Automatic Step Tracking** – STEP-UP will utilize the smartphone's built-in motion sensors to automatically count steps in real time without requiring additional devices or manual input. 6. **Activity Summary and Progress Monitoring** – The application will provide daily and weekly summaries of step performance, including progress bars and milestone indicators to help users monitor their physical activity levels. 7. **Gamification and Reward System** – Accumulated steps will be converted into in-game points that users can collect and use for rewards. This gamified system is designed to increase motivation and make walking more engaging. 8. **Daily Goals and Weekly Missions** –The application will implement a milestone-based daily step progression system (2,500 to 11,000 steps) aligned with established physical activity classifications. In addition, optional weekly missions based on campus attendance (four days per week) will encourage step accumulation, consistency, and gradual improvement to support sustainable physical activity behavior. The mission structure is designed to be modular, allowing future enhancements without altering the core system functionality. 9. **Campus-Based Map Visualization** – The application will include a 2D bird's-eye view map of the university campus where users can view their avatar and walking progress, increasing engagement with the campus environment. 10. **Augmented Reality (AR) Walking Mode** – STEP-UP will feature an AR mode that uses the smartphone camera to display the user's avatar and step progress in real

	time, making walking more interactive and immersive. **11. Avatar Customization** – Users will be able to personalize their in-app avatar by purchasing outfits and accessories using earned points. This feature increases engagement and provides additional incentives to remain active. **12. Health Guidance Features** – The application will include posture tutorials, warm-up and cooldown guides, and general fitness tips to promote safe and proper physical activity habits. **13. BMI and Estimated Calories Burned** – Users can input their height and weight to compute their Body Mass Index (BMI) and receive estimated calorie expenditure based on daily step counts. **14. Vehicle Detection Anti-Cheat Mechanism** – The system will implement a coded speed threshold detection mechanism that pauses step counting when movement exceeds a predefined speed (e.g., 10 km/h) to prevent inaccurate step recording during vehicle use. **15. Safety Reminders** – The application will display preventive pop-up notifications that encourage users to stay aware of their surroundings and walk responsibly within campus areas. These reminders are precautionary only and do not include real-time hazard detection. **16. Leaderboard** – The STEP-UP application will include a leaderboard feature that ranks users based on their accumulated steps and earned points within a selected time period (e.g., daily or weekly). This feature encourages healthy competition among students and increases motivation to remain physically active. Rankings will display usernames to maintain user privacy, and data will be synchronized through Firebase to ensure updated results.
Project Workplan	Please see the attached form below.
Sustainability Plan	The sustainability of the STEP-UP: Augmented Reality Mobile Application for Promoting Physical Activity Among University Students will be ensured through proper maintenance, continuous improvement, and long-term usability planning. The following strategies outline how the project will remain functional and beneficial beyond its initial development phase: **1. Maintenance and Operational Continuity** The STEP-UP application supports offline functionality through local data storage; however, cloud-based features such as leaderboard synchronization and data backup require internet connectivity. This hybrid design ensures both accessibility and scalability while maintaining operational continuity. **2. Software and Feature Updates** The application will be developed using organized and well-documented coding

practices to allow future updates and enhancements. The system structure will enable new features to be added without requiring major changes to the existing application. Possible future improvements may include expanded gamification mechanics, additional augmented reality features, improved user interface design, and integration with other health-related applications. These potential upgrades will help keep the application relevant and useful over time.

3. Continuous Evaluation and Improvement

User feedback and evaluation results gathered during the testing phase will serve as the primary basis for further system enhancement. The application can undergo continuous evaluation by future student researchers or developers to assess its effectiveness in promoting physical activity. Suggestions from users and health professionals can be incorporated to improve usability, engagement, and overall system performance.

4. Documentation and Knowledge Transfer

To ensure the long-term sustainability of the project, complete documentation will be prepared, including system manuals, source codes, design files, and user guides. These materials will be properly organized and turned over to the institution or future capstone groups. This knowledge transfer process will allow the application to be maintained, upgraded, or expanded by succeeding developers without difficulty.

5. Institutional and Community Relevance

The core purpose of STEP-UP—promoting physical activity among university students—addresses an ongoing and relevant need within the campus community. Because student health and wellness will always be a priority, the application can continue to be utilized as a tool for encouraging active lifestyles. The system may also be adopted by university programs or wellness initiatives that aim to support student well-being.

6. Scalability and Future Expansion

The application has the potential to be expanded beyond its initial implementation. Future versions may include features such as online leaderboards, social interaction modules, multi-campus maps, or integration with wearable devices. These enhancements can increase user engagement and extend the usefulness of the system to a larger audience.

Through proper maintenance, continuous evaluation, organized documentation, and the possibility for future enhancements, the STEP-UP application is expected to remain a functional, relevant, and sustainable system. These sustainability strategies ensure that the project will continue to benefit university students and support the promotion of physical activity even after the completion of the capstone research.

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PROJECT OBJECTIVES	TARGET ACTIVITIES	OBJECTIVELY VERIFIABLE INDICATORS/TARGET ACCOMPLISHMENTS	1	2	3	4	5	6	7	8
1. Foundation & Structure **To establish the app navigation, local storage,** **And cloud syncing architecture.**	Implementation of Layer 0.1: Main Menu / Navigation Panels Implementation of Layer 0.2: Local Save System Implementation of Layer 0.3: Firebase Offline – online Syncing	App navigates between screens smoothly User data Saves Even after closing the app Data Syncs to cloud upon reconnection.								
2. Core Tracking **To implement the fundamental health monitoring features.**	Implementation of Layer 1.1: Automatic Step tracking sensor Implementation of Layer 1.2: Activity Summary UI & Progress Bars.	Accurate Step count displayed on User Interface Daily Progress bar updates in real – time, and calibrating sensor activity								
3. Gamification & Map (Layers 2 & 3) **Adding the game economy and campus visualization.**	Implementation of Layer 2.1: Step to point Conversion. Implementation of Layer 2.2: Daily Mission System. Implementation of Layer 3.1:	Points increase relative to steps taken. Daily missions reset every 24 hours.								

	PSAU 2D Bird's eye map Integration.	User market moves correctly on the 2D plain Map.								
4. AR & Avatar System (Layer 4 & 5) **Integrating the Augmented reality and user retention rewards.**	Implementation of Layer 4.1: AR Camera – Based Walking View. Implementation of Layer 5.1: Avatar Customization & Shop System. Integrationg AR Camera permission handling	Camera Feed active with UI overlays. Users can purchase items with points. Selected avatar items appear in game.								
5. Health Guide & Safety (Layers 6 & 7)	Implement	User satisfaction feedback with key areas for improvement.								
6. Review and Updation	Implement user feedback Fix identified bugs Improve UI/UX design Enhance forecasting accuracy	Improved prototype with refined features and better accuracy.								
7. Final Implementation & Deployment	Deploy to production server Configure security & backups Conduct user training Final system testing	Fully functional system deployed and ready for use. Results documented.								

	Write results & discussion chapter									
8. Documentation & Paper Writing	Complete system documentation Write introduction & conclusion Finalize research paper Prepare defense presentation Submit final paper	Complete research paper and system documentation submitted.								